Analytic Positivity of the Aggregate Base Derivative

For $0 \le s \le 1$, set

$$G_1(s) = \frac{\sqrt{1-s^2} - s\arccos s}{\pi}.$$

For $0 < \rho < 1$ and $K > 0$, put

$$\eta_\rho = G_1(\max\{\rho, 1/2\}), \qquad d_\rho = \frac{2\arcsin\rho}{\pi}, \qquad b_\rho = \frac{2\pi\rho}{1-\rho},$$

$$\mu = \rho K, \qquad \overline{R} = \mu + \frac{1}{2}, \qquad S = \sqrt{\overline{R}^2 + b_\rho K},$$

$$I(\rho, K) = \frac{1}{2}\left[\overline{R}S + b_\rho K \operatorname{arsinh}\frac{\overline{R}}{\sqrt{b_\rho K}} - \overline{R}^2\right],$$

and define the continuous aggregate reserve

$$\begin{aligned}\mathcal{B}(\rho, K) = (\mu - \tfrac{1}{2})(K\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - \tfrac{3}{2})(\mu - \tfrac{1}{2})\\ - (1 + d_\rho)I(\rho, K) - \frac{8(\mu + \frac{1}{2})}{15\sqrt{\mu}}.\end{aligned} \tag{1}$$

Direct differentiation gives

$$I_K = \rho(S - \overline{R}) + \frac{b_\rho}{2}\operatorname{arsinh}\frac{\overline{R}}{\sqrt{b_\rho K}}, \tag{2}$$

and

$$\begin{aligned}\mathcal{B}_K = \rho(K\eta_\rho - 1) + (\mu - \tfrac{1}{2})\eta_\rho + d_\rho\rho(\mu - 1)\\ - (1 + d_\rho)I_K - \frac{2\rho(2\mu - 1)}{15\mu^{3/2}}.\end{aligned} \tag{3}$$

Lemma 1 (Analytic positivity of the aggregate base derivative)**.** *For*

$$\frac{7}{51} \le \rho \le \frac{39}{50},$$

the continuous aggregate reserve defined in (1) *satisfies*

$$\mathcal{B}_K(\rho, 200) > 0.$$

This assertion uses no numerical or interval certificate.

Proof. We use only the elementary rational bounds

$$3 < \pi < \frac{22}{7}, \qquad \frac{78}{25} < \pi, \qquad \frac{7}{5} < \sqrt{2}, \qquad \frac{265}{153} < \sqrt{3} < \frac{26}{15}. \tag{4}$$

The two square-root bounds in (4) follow immediately by squaring; the displayed rational bounds for π are the classical Archimedean bounds.

Put

$$\mu = 200\rho, \qquad b_\rho = \frac{2\pi\rho}{1-\rho}, \qquad \overline{R} = 200\rho + \frac{1}{2}, \qquad S = \sqrt{\overline{R}^2 + 200b_\rho},$$

and

$$u = \frac{\overline{R}}{\sqrt{200b_\rho}}, \qquad P = S - \overline{R}.$$

We first obtain a uniform elementary upper bound for the derivative of the integral term. Since $\rho > 1/8$, one has $\overline{R} < 204\rho$. Therefore

$$u^2 < \frac{204^2\rho(1-\rho)}{400\pi} \leq \frac{41616}{1600\pi} < \frac{867}{100} < 9.$$

Thus $u < 3$. Moreover

$$\sinh 2 = 2 + \frac{2^3}{3!} + \frac{2^5}{5!} + \cdots > \frac{10}{3} > 3,$$

and hence $\operatorname{arsinh} u < 2$. The identity

$$P = \frac{200b_\rho}{S + \overline{R}}$$

and the inequalities $S > \overline{R} > 200\rho$ give $P < b_\rho/(2\rho)$. Formula (2) now yields

$$I_K(200) = \rho P + \frac{b_\rho}{2}\operatorname{arsinh} u < \frac{3b_\rho}{2}. \tag{5}$$

The final error term in (3) obeys

$$0 < \frac{2\rho(2\mu - 1)}{15\mu^{3/2}} < \frac{4\sqrt{\rho}}{15\sqrt{200}} < \frac{2}{105} < \frac{1}{50}. \tag{6}$$

At $K = 200$, formula (3) becomes

$$\begin{aligned} \mathcal{B}_K(\rho, 200) = 400\rho\eta_\rho - \rho - \frac{\eta_\rho}{2} + d_\rho\rho(200\rho - 1) \\ - (1 + d_\rho)I_K(200) - \frac{2\rho(400\rho - 1)}{15(200\rho)^{3/2}}. \end{aligned} \tag{7}$$

We first treat $7/51 \leq \rho \leq 1/2$. On this interval

$$\eta_\rho = \omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \qquad 0 < d_\rho \leq \frac{1}{3}.$$

The last two bounds in (4) give

$$\frac{13}{120} < \eta_\rho < \frac{1}{9}.$$

Indeed,

$$\frac{\sqrt{3}}{2\pi} > \frac{265}{153}\frac{7}{44} = \frac{1855}{6732} > \frac{11}{40},$$

whereas

$$\frac{\sqrt{3}}{2\pi} < \frac{26}{15}\frac{25}{156} = \frac{5}{18}.$$

The term $d_\rho \rho(200\rho - 1)$ is nonnegative. Combining (5), (6), $2\pi < 44/7$, and (7), we obtain

$$\begin{aligned}\mathcal{B}_K(\rho, 200) &> \rho\left(\frac{127}{3} - \frac{88}{7(1-\rho)}\right) - \frac{17}{225} \\ &\geq \frac{7}{51}\left(\frac{127}{3} - \frac{176}{7}\right) - \frac{17}{225} = \frac{2912}{1275} > 0.\end{aligned}$$

It remains to treat $1/2 \leq \rho \leq 39/50$. Monotonicity of arcsin, together with $\arcsin \rho \geq \rho$, gives

$$\frac{7\rho}{11} < d_\rho < \frac{3}{5}. \tag{8}$$

For the upper bound, note that

$$\frac{39}{50} < \sin\frac{3\pi}{10} = \frac{1+\sqrt{5}}{4};$$

the strict rational comparison follows from $\sqrt{5} > 53/25$. For the lower bound use $2/\pi > 7/11$.

Since

$$\eta_\rho = G_1(\rho) = \frac{1}{\pi}\int_\rho^1 \arccos t \, dt$$

and $\arccos t \geq \sqrt{2(1-t)}$, we have

$$\eta_\rho \geq \frac{2\sqrt{2}}{3\pi}(1-\rho)^{3/2} > \frac{49}{165}(1-\rho)^{3/2}. \tag{9}$$

Also $0 < \eta_\rho \leq \omega_0 < 1/9$.

Let $[a, c]$ be any of the six intervals in the table below, and let s_c be the displayed rational number. Direct squaring proves $s_c^2 \leq 1 - c$. Hence (9) gives

$$\eta_\rho \geq \eta_c := \frac{49}{165}(1-c)s_c \qquad (a \leq \rho \leq c).$$

Equations (5)–(8) then imply the entirely rational lower bound

$$\mathcal{B}_K(\rho, 200) > L(a, c), \tag{10}$$

where

$$L(a,c) := 400a\eta_c - c - \frac{1}{18} + \frac{7a^2}{11}(200a - 1) - \frac{528}{35}\frac{c}{1-c} - \frac{1}{50}.$$

Every entry in the last column is obtained by substitution and integer cross-multiplication:

a	c	s_c	η_c	$L(a,c)$
$\frac{1}{2}$	$\frac{11}{20}$	$\frac{2}{3}$	$\frac{49}{550}$	$\frac{251291}{17325}$
$\frac{11}{20}$	$\frac{3}{5}$	$\frac{5}{8}$	$\frac{49}{660}$	$\frac{70619}{5040}$
$\frac{3}{5}$	$\frac{13}{20}$	$\frac{7}{12}$	$\frac{2401}{39600}$	$\frac{576451}{44100}$
$\frac{13}{20}$	$\frac{7}{10}$	$\frac{6}{11}$	$\frac{147}{3025}$	$\frac{4940821}{435600}$
$\frac{7}{10}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{49}{1320}$	$\frac{2411}{315}$
$\frac{3}{4}$	$\frac{39}{50}$	$\frac{23}{50}$	$\frac{1127}{37500}$	$\frac{11101801}{1386000}$

All six rational lower bounds are strictly positive, and the intervals tile $[1/2, 39/50]$. This proves the assertion on the high branch and completes the proof. $\square$